Supplementary Material

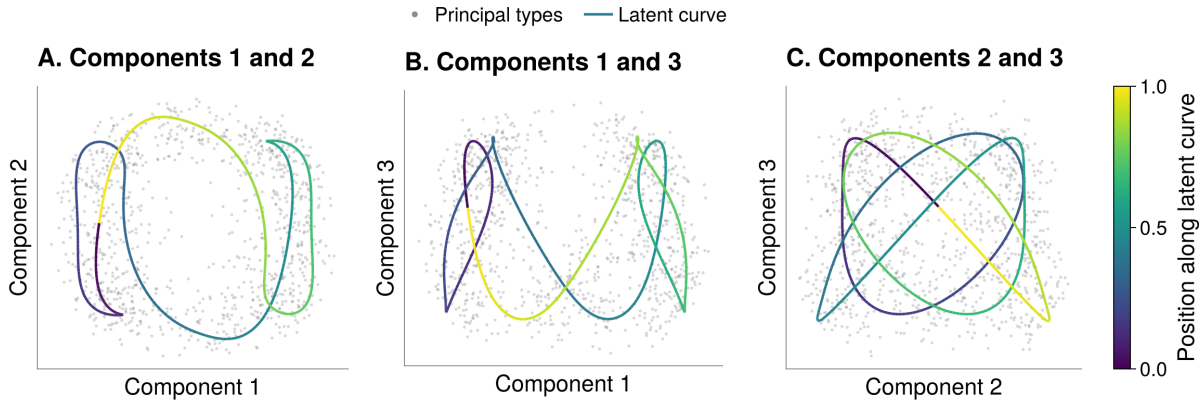


Figure S1: Principal types and their latent curve. All 1000 baseline types from realization 1, projected onto pairs of the curve's first three principal components (A–C). Gray points are realized types; the line shows the latent curve, colored by position along it. Matching and learning use the full eight-dimensional types, not these projections.

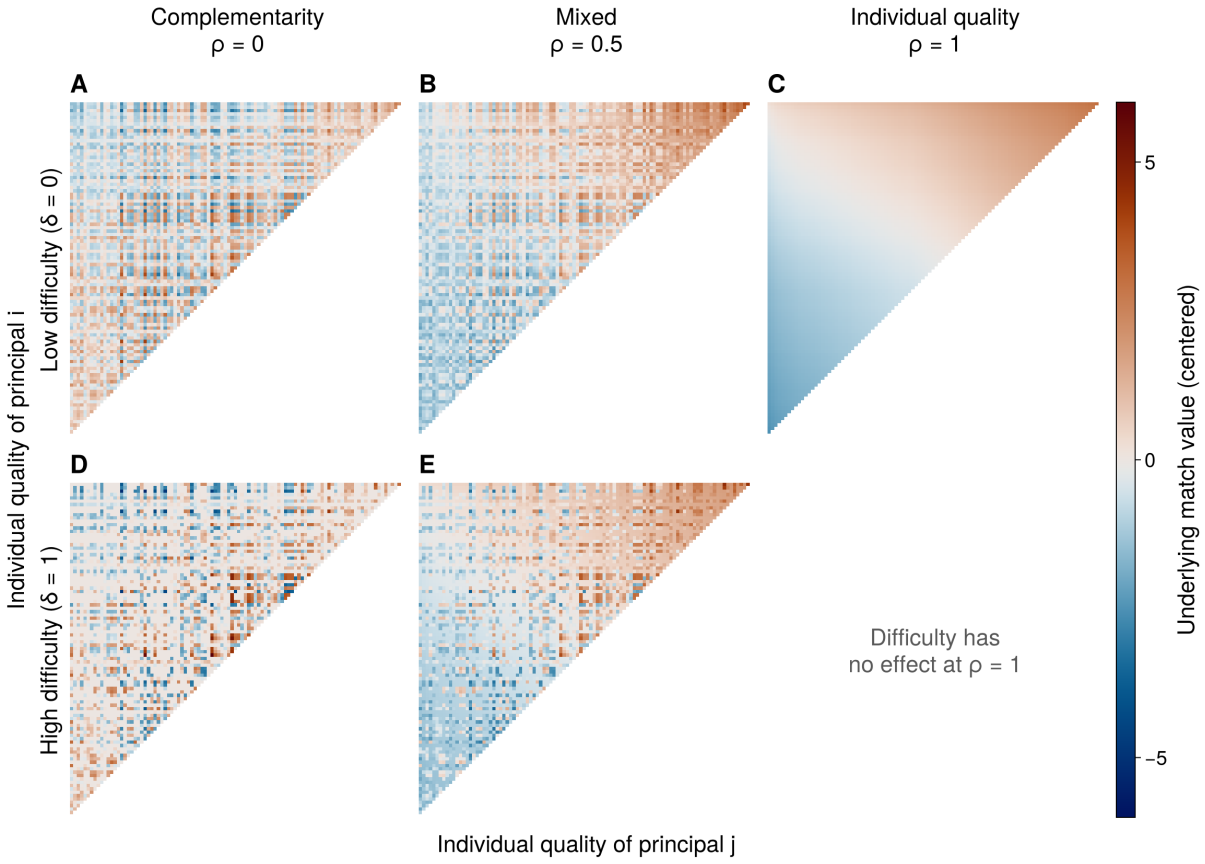


Figure S2: Underlying match value across pairs of principal types. 100 types selected at evenly spaced ranks of individual quality from realization 1 (1000 principals). Types and matching-function objects c, A, and B are fixed across panels. Individual quality increases rightward and upward. Upper triangles exclude self-matches; colors show each pair's underlying value minus the grand mean of the full value matrix.

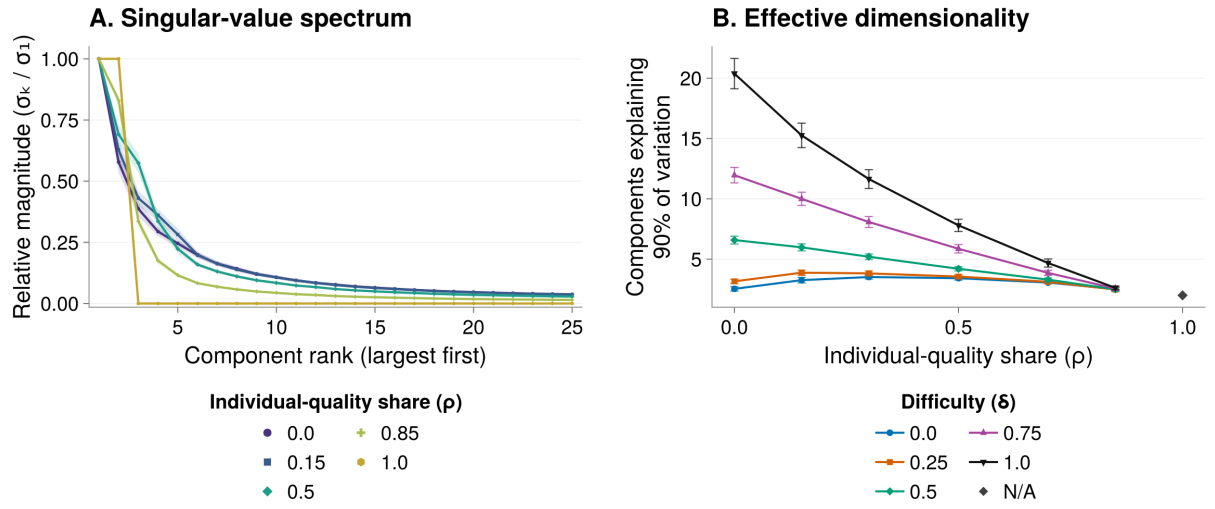


Figure S3: Dimensionality of underlying match value. Matrices of pair-specific underlying match values, centered within each realization and summarized over 50 independent realizations. Types and matching-function draws are fixed within each realization as ρ and δ vary. A: the first 25 singular values at $\delta = 0.5$, divided by the largest within each realization. B: the fewest components accounting for 90% of the sum of squared singular values. Lines and markers are means; ribbons and whiskers show pointwise 95% Monte Carlo intervals. N/A denotes $\rho = 1$, where underlying match value depends only on individual quality and difficulty is inapplicable.

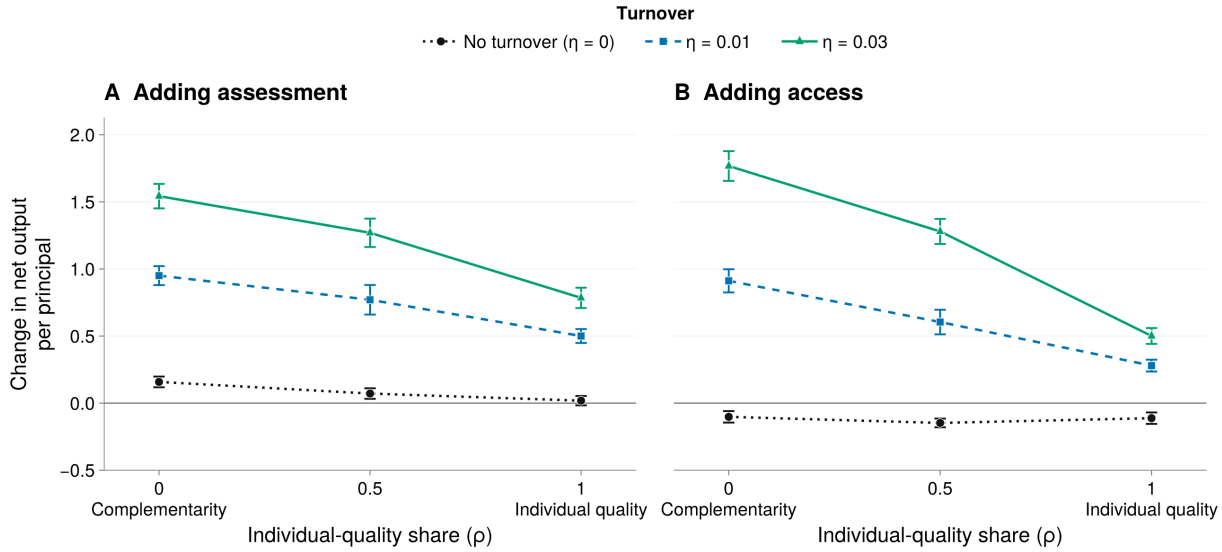


Figure S4: Contributions of assessment and access across matching composition. Late-mean differences paired by seed: A, full service minus access-only; B, full service minus assessment-only. Parameters other than matching composition and turnover remain at baseline. Points average 20 seeds per regime; bars show pointwise 95% Monte Carlo intervals. Connecting lines are visual guides.

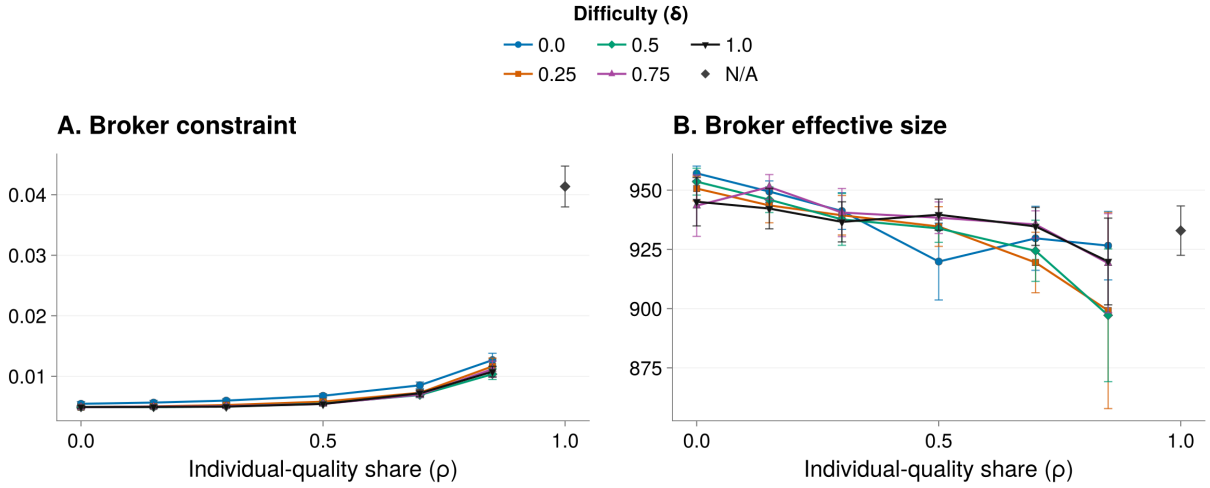


Figure S5: Alternative structural measures across the matching problem. Burt's constraint measures the concentration of the broker's direct and indirect ties; effective size discounts contacts tied to one another. Points show ensemble late means across the $\rho \times \delta$ design, with other parameters at baseline. Whiskers show pointwise 95% Monte Carlo intervals. N/A denotes $\rho = 1$, where underlying match value depends only on individual quality and difficulty is inapplicable.

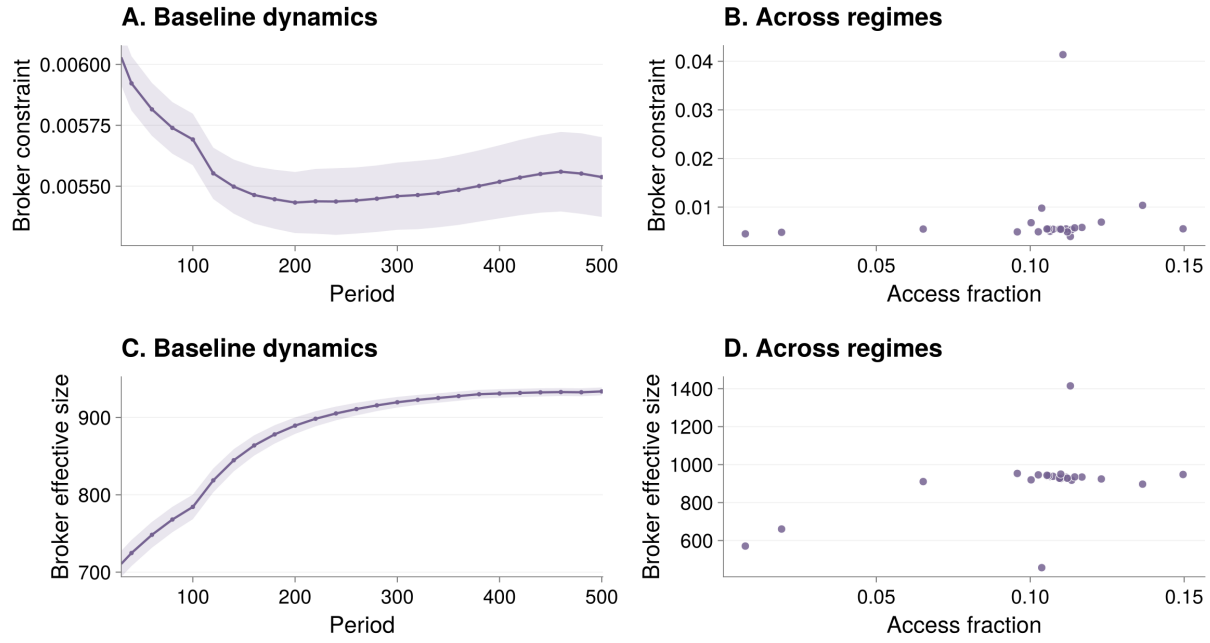


Figure S6: Alternative structural measures over time and against access. A and C: baseline measures recorded every 20 periods and smoothed within each seed using a 5-observation rolling mean, shown from $t = 30$. Lines are ensemble means; ribbons show pointwise 95% Monte Carlo intervals. B and D: ensemble late means for the 26 regimes that vary one parameter at a time from baseline.

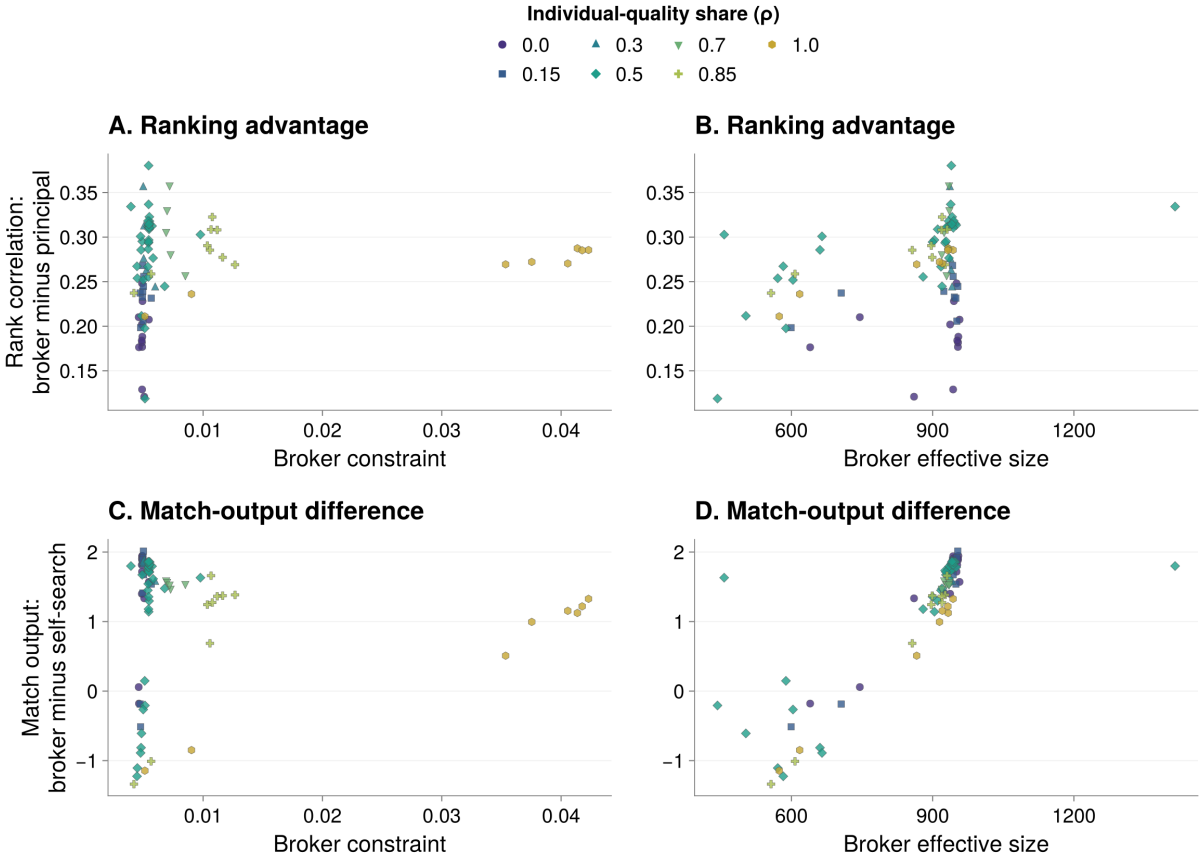


Figure S7: Ranking and output differences by alternative structural measures. Points show ensemble late means for 80 regimes. C–D compare mean realized output per completed match (brokered minus self-search), before costs and fees.